THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

December 14, 2016 9:30am - 12:00noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer All **SEVEN** Questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (*20 points*) For each of the questions below there is only ONE correct answer. Write your answer in the answer book provided. No steps/explanations are required.

(1) Which of the following is/are TRUE for all subsets X, Y, Z of a universal set U?

I. $X \times Y = Y \times X$;

II. $(X \cap Y) \times Z = (X \times Z) \cap (Y \times Z)$;

III. $X \backslash (Y \cap Z) = (X \backslash Y) \cup (X \backslash Z)$.

A. II only; B. I and III only; C. II and III only; D. I, II and III.

(2) Let $P(x)$ and $Q(x)$ be statements concerning elements x in some set S. Which of the following statements use "$\Longrightarrow$" correctly?

I. $(\exists x \in S,\, P(x)) \wedge (\exists x \in S,\, Q(x)) \Longrightarrow (\exists x \in S,\, P(x) \wedge Q(x))$;

II. $(\exists x \in S,\, P(x) \wedge Q(x)) \Longrightarrow (\exists x \in S,\, P(x)) \wedge (\exists x \in S,\, Q(x))$;

III. $(\exists x \in S,\, P(x) \vee Q(x)) \Longrightarrow (\exists x \in S,\, P(x)) \vee (\exists x \in S,\, Q(x))$.

A. II only; B. III only; C. II and III only; D. I, II and III.

(3) Which of the following is the contrapositive of the statement "A finite set is countable."?

I. A countable set is finite;

II. Some countable sets are not finite;

III. An uncountable set is infinite.

A. I; B. II; C. III; D. None of these.

(4) Which of the following is the converse of the statement "An uncountable set is infinite."?

I. A countable set is not necessarily infinite;

II. An infinite set is uncountable;

III. There are uncountable sets.

A. I; B. II; C. III; D. None of these.

(5) Which of the following is/are the negation of the statement that "$\lim_{n\to\infty} a_n = 2$"?

I. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $|a_n - 2| < \varepsilon$ for all $n \geq N$;

II. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $|a_n - 2| \geq \varepsilon$ for all $n \geq N$;

III. $\exists \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $|a_n - 2| < \varepsilon$ for all $n \geq N$;

IV. $\exists \varepsilon > 0$ such that $\forall N \in \mathbb{N}$, $\exists n \geq N$ such that $|a_n - 2| \geq \varepsilon$.

A. I and II only; B. II only; C. III only ; D. IV only.

2. (*10 points*) For an integer $n \geq 0$, give the necessary and sufficient condition for $2^n + 1$ to be divisible by 3 and explain your work.

3. (*10 points*) Prove or disprove the following statements:

   1) For any integers a, b and positive integer c, if $a|c$ and $b|c$, then $ab|c$;

   2) There exist integers a and b such that $35a + 17b = 283$.

4. (*10 points*) Consider the map $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f(x, y) = \left(\pi x + \sqrt{2}y,\ x - \sqrt{2}y\right).$$

   Assuming that π is irrational. Show that $(\mathbb{Q}_{\neq 0} \times \mathbb{R}) \cap f^{-1}(\mathbb{Q}^2) = \emptyset$, where $\mathbb{Q}$ is the set of all rational numbers, and $\mathbb{Q}_{\neq 0}$ is the set of all non-zero rational numbers.

5. (*15 points*) Consider the relation on $\mathbb{R}$ defined by

$$x \sim y \quad \text{if and only if} \quad \exists q \in \mathbb{R},\ q > 0, \quad \text{such that } x = qy.$$

   1) Prove that this relation is an equivalence relation;

   2) Describe all the equivalence classes of $\mathbb{R}$ under $\sim$ and the partition of $\mathbb{R}$ defined by $\sim$; Find the cardinality of the quotient set $\mathbb{R}/\sim$.

6. (*15 points*) 1) Let A be any set and let $P(A)$ be the power set of A. Let $f : A \to P(A)$ be any map, and let

$$B = \{a \in A : a \notin f(a)\} \in P(A).$$

   Prove that B is not in the image of f;

   2) Prove that $P(\mathbb{Q})$ is uncountable, where $\mathbb{Q}$ is the set of rational numbers

7. (*20 points*) The sequence $a_1, a_2, a_3, \ldots$ is defined by $a_1 = 2$ and $a_{n+1} = a_n^2 - a_n + 1$ for all $n \geq 1$. Define

$$S_n = \sum_{k=1}^{n} \frac{1}{a_k} = \frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}, \qquad n \geq 1.$$

   1) Compute a_2, a_3, a_4, a_5 and S_1, S_2, S_3 and S_4;

   2) Show that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{a_n}$ is convergent and find the sum. Explain all the steps.

END OF PAPER